SAMPLE

Spacetime

The Fabric of the Universe

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Introduction

Somewhere between the chair you are sitting on and the clock on the wall, modern physics found a seam. For more than two centuries, scientists treated space as a stage and time as a metronome: the stage held the actors, the metronome kept a beat that everyone in the theater heard identically, and neither the stage nor the metronome had any influence on the play. Around 1905 that comfortable arrangement collapsed. Within a few years, physicists stopped speaking of space and time as separate things and began speaking of a single four-dimensional fabric that carries both at once. This book is about that fabric: what it is, where the idea came from, what it can do, and where it still resists understanding.

The fusion did not arrive as a single announcement. It crept in through a patent clerk's thought experiments about light, crystallized in a mathematician's lecture that his own students reportedly found impenetrable, and grew teeth when it turned out to explain a wobble in Mercury's orbit that astronomers had chased for half a century. Along the way it dissolved simultaneity, stretched and slowed clocks, bent starlight, and handed engineers a correction factor without which satellite navigation would drift by kilometers per day. Spacetime is one of the rare ideas that is simultaneously the backbone of modern physics and a working tool in ordinary pocket electronics.

A word about what this book is not. The general theory of relativity, its companion volume in this series, walks through curved geometry in loving detail. The special theory has its own volume too, built around moving frames and the curious behavior of fast travelers. Here the subject is the arena itself. When the story needs a result from those siblings, I will state it plainly, show why it matters to the fabric, and point you toward the deeper treatment rather than repeating it.

The journey runs as follows. The first chapters look backward: how thinkers from Aristotle through Newton to Mach argued about whether space and time are real things or mere relations among things, and how experiments with light at the close of the nineteenth century made the old answers untenable. The middle chapters enter the fused arena: the interval that all observers agree on, the light cones that divide "can affect" from "cannot affect," and the worldlines that replace particles with patient four-dimensional threads. The later chapters ask what happens when the fabric bends, ripple, tears, and stretches: the tests that confirmed it, the waves that shook it, the holes in it, and the cosmic expansion written directly into its geometry. The closing chapters confront the frontier: the stubborn refusal of spacetime to sit politely with quantum mechanics, and the disquieting possibility that the arena is itself an emergent thing, like temperature, rather than a fundamental thing, like an atom.

No equations await you. The ideas are geometric, and geometry can be drawn in words. What awaits you instead is a single guiding image, offered now and repeated throughout: the universe as a vast, four-dimensional loaf, and every object in it, from mayflies to galaxies, as a thread drawn through the loaf from its beginning to its end. Read on, and the loaf will start to feel real.

The history that this chapter recounts has a habit of being told as though it were a mystery solved in one afternoon, and the habit deserves correcting at the outset. The tension between the two rooms was noticed for decades before anyone resolved it, and the resolution, when it came, was resisted by its own author longer than popular accounts admit. The fused fabric was not invented in a burst of intuition but assembled slowly, from failed experiments, a dismissed ether, a clerk's rereading of simultaneity, and a professor's mathematical patience. What follows is the story in that slower, truer tempo, and the reader is asked to hold one question in mind throughout: not what the fabric is, which the later chapters must answer, but what it took to notice that the question existed at all.

Chapter 1: Two Rooms and an Argument

Imagine two rooms. In one, a tall, solitary figure insists that space and time are absolute: a container exists whether or not anything fills it, and duration flows whether or not anyone measures it. In the other room, a smaller man with a sharp wit replies that this is nonsense, that space is only the order of bodies and time only the order of events, and that talking about empty containers is talking about nothing at all. Isaac Newton and Gottfried Leibniz conducted their quarrel by letter, through intermediaries, in the 1710s, and every argument about the nature of space and time since has been a footnote to it.

Newton had practical reasons for his position. His mechanics worked by describing motion against a fixed background. Acceleration, he argued, has to be acceleration relative to something absolute, otherwise a spinning bucket of water has nothing to press its surface into a curve against. His famous bucket hung by a rope: twist the rope, let go, and the water climbs the walls. Relative to the bucket, the water is at rest at first, yet its surface

curves. Something beyond the bucket, beyond the room, beyond every object in the universe, must supply the standard against which the water turns. Newton named it absolute space, and gave time the same dignity: true time, he wrote, flows equably without relation to anything external.

Leibniz replied with a razor that philosophers still wield. If space is a real container, then the universe could have been created a mile east of where it sits, or a day earlier, and these would be different universes. But what would make them different? Every object would bear the same relations to every other object. God, whom both men took seriously as an economist of reasons, would have no ground for choosing. Since a choice without a reason is beneath a rational creator, space and time cannot be containers; they must be systems of relations among the things that exist. Time is not a river; it is the order of happenings, as a genealogy is not a ladder in a library but the order of family members.

Newton won the physics of the next two centuries, partly because his mathematics ran and Leibniz's metaphysics merely stood. Engineers measured distances and durations and never once needed to ask whether the distance was "really" there when nobody looked. The bucket argument in particular haunted anyone who tried to dismiss absolute space entirely: acceleration, unlike constant velocity, produces felt effects, and the felt effects demand a background.

Yet the cracks accumulated. In the 1880s Ernst Mach, an Austrian physicist with a talent for unsettling everybody, published his criticisms. The water in Newton's bucket, he suggested, climbs the walls because it rotates relative to the distant masses of the universe, not relative to an invisible container. Remove everything but the bucket, and the notion of its spinning loses meaning. The background, if it exists, is the stars. A young reader in Bern would later call this book-reading episode a major jolt to his thinking, and the idea that local physics depends on the whole universe would echo through general relativity, where the geometry of the stage is set by everything standing on it.

The second crack came from light. By the late 1800s, Maxwell's equations described light as a wave rippling through an all-pervading medium, the luminiferous ether, which would supply exactly the sort of absolute frame Newton wanted. If the earth plows through the ether on its orbit, light should race past us faster when we swim with the current than against it. Albert Michelson and Edward Morley built the most sensitive interferometer of their age to catch the difference. Their instrument could have detected a drift far smaller than the orbital motion predicted. It found nothing. The current did not exist. The stage, which the nineteenth century had finally furnished with a physical medium, turned out to be empty, and a stage that is nothing but emptiness is a stage that carries a question mark.

By 1900, then, the argument in the two rooms was unresolved but freshly armed. Newton's absolute background was needed by mechanics and disowned by experiment. Leibniz's relational view was philosophically elegant and physically vague. Nature, as it turned out, had a third answer prepared, one that neither disputant could have phrased: the background is real, but it is neither space alone nor time alone, and its geometry is not the one Euclid drew.

The ether deserves a fuller obituary than the textbook usually gives it, because the idea was not foolish. Light behaved, unmistakably, like a wave, and every wave anyone had ever met lived in some medium: sound in air, ripples in water, tremors in rock. To postulate a medium for light was to extend a perfectly successful pattern of reasoning, not to indulge a fantasy. The failure came only at the level of detail, and it came strangely: the medium had to be stiff enough to carry the fastest known signal and yet offer no resistance to the planets sweeping through it, rigid as steel and thin as nothing. It is worth honoring the ether as the best available idea of its time, if only because the lesson generalizes; the vacuum that replaced it, as the later chapters will show, turned out to be no simple emptiness either, and the fabric's final portrait may hold surprises as unmannerly as the ether's did.